

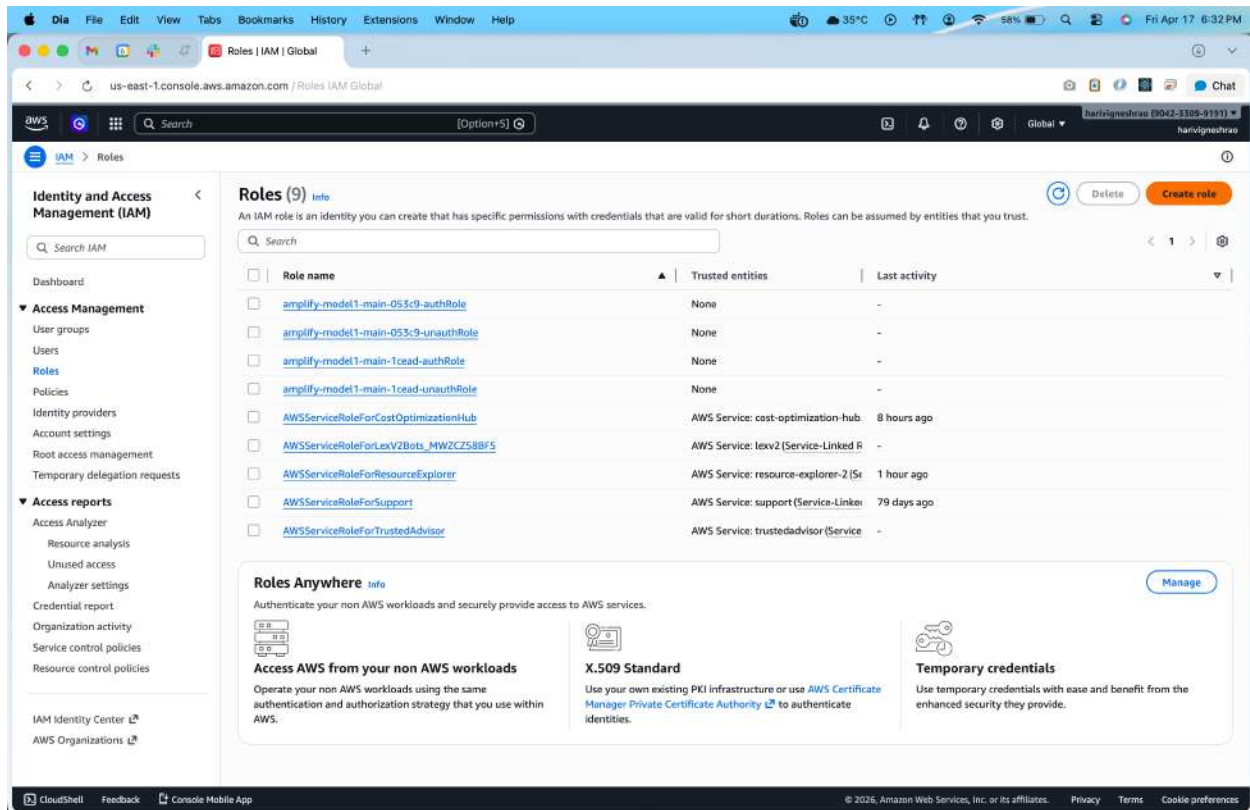
HARI VIGNESH RAO

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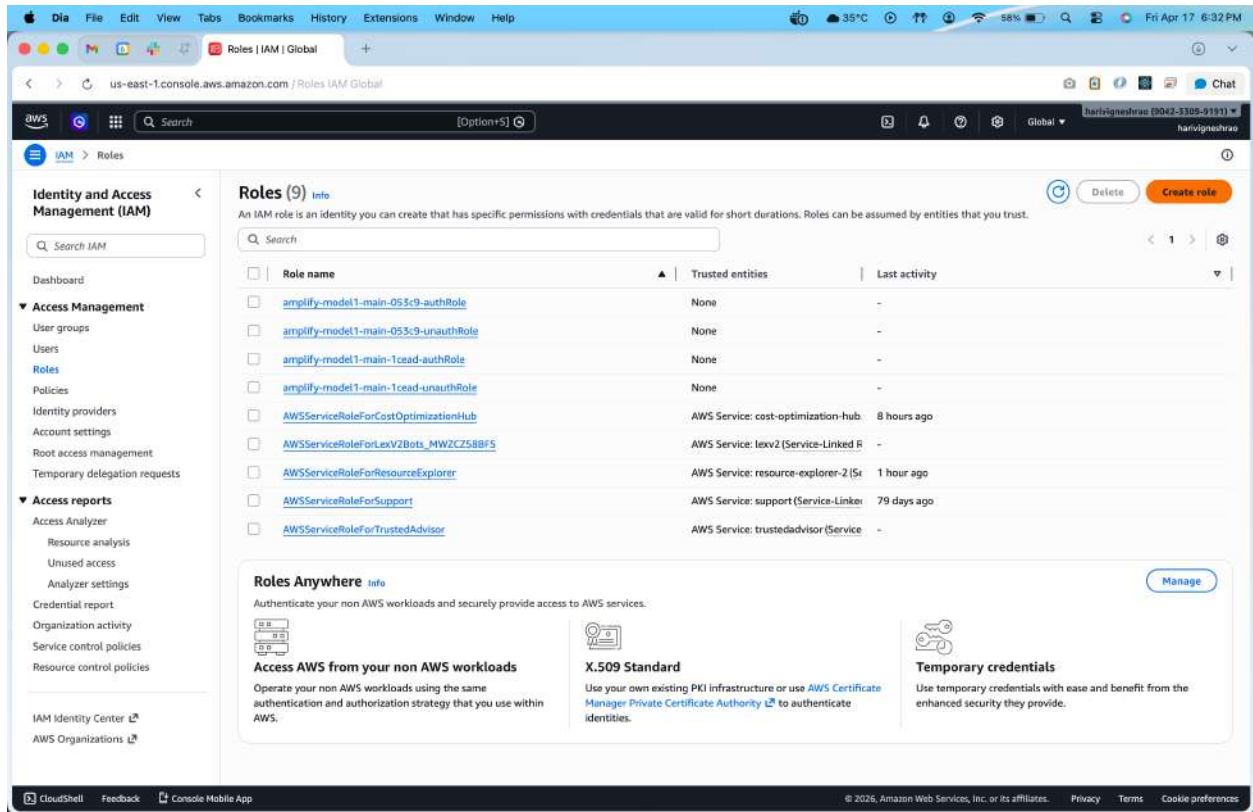
Procedure: Attach an IAM Role to an EC2 Instance (S3 access without keys)

Step 1 — Create Role in Amazon Web Services IAM

1. Open IAM Console → Roles → Create role

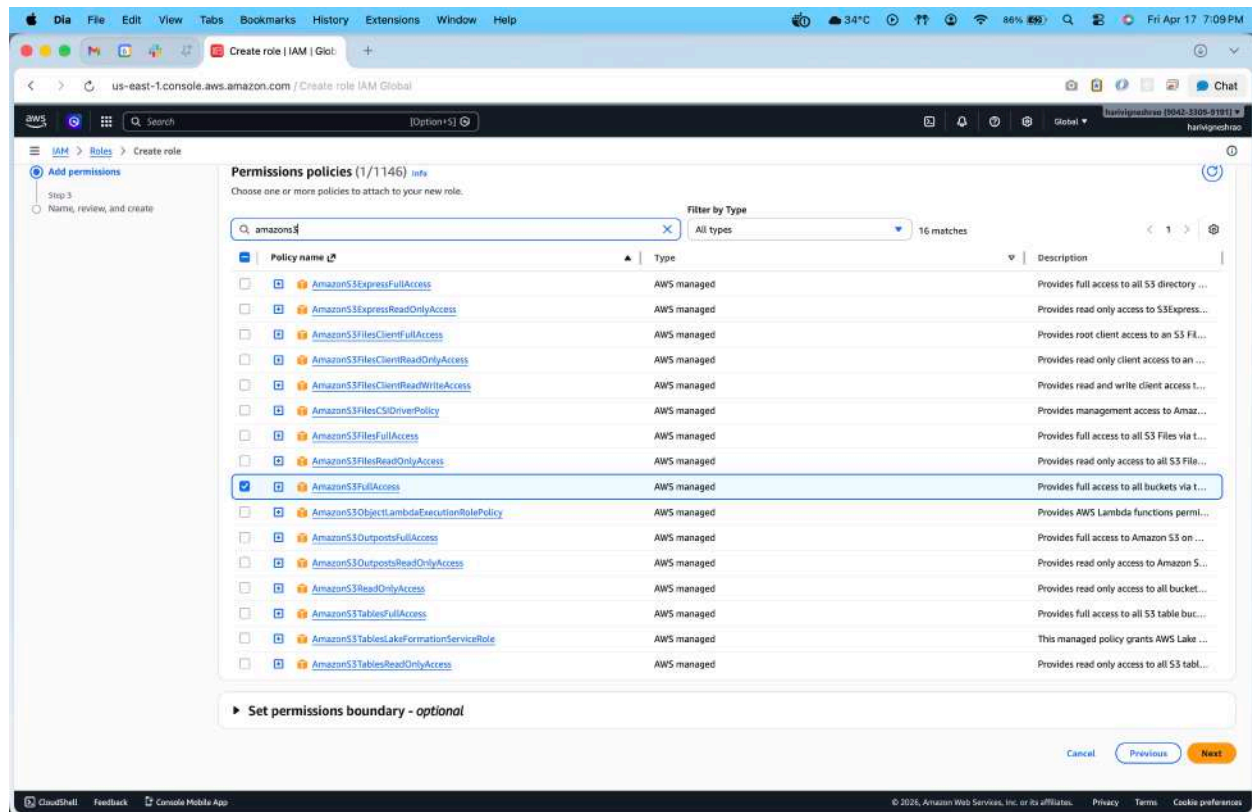


2. Trusted entity: Select AWS Service
3. Use case: Choose EC2
4. Click Next



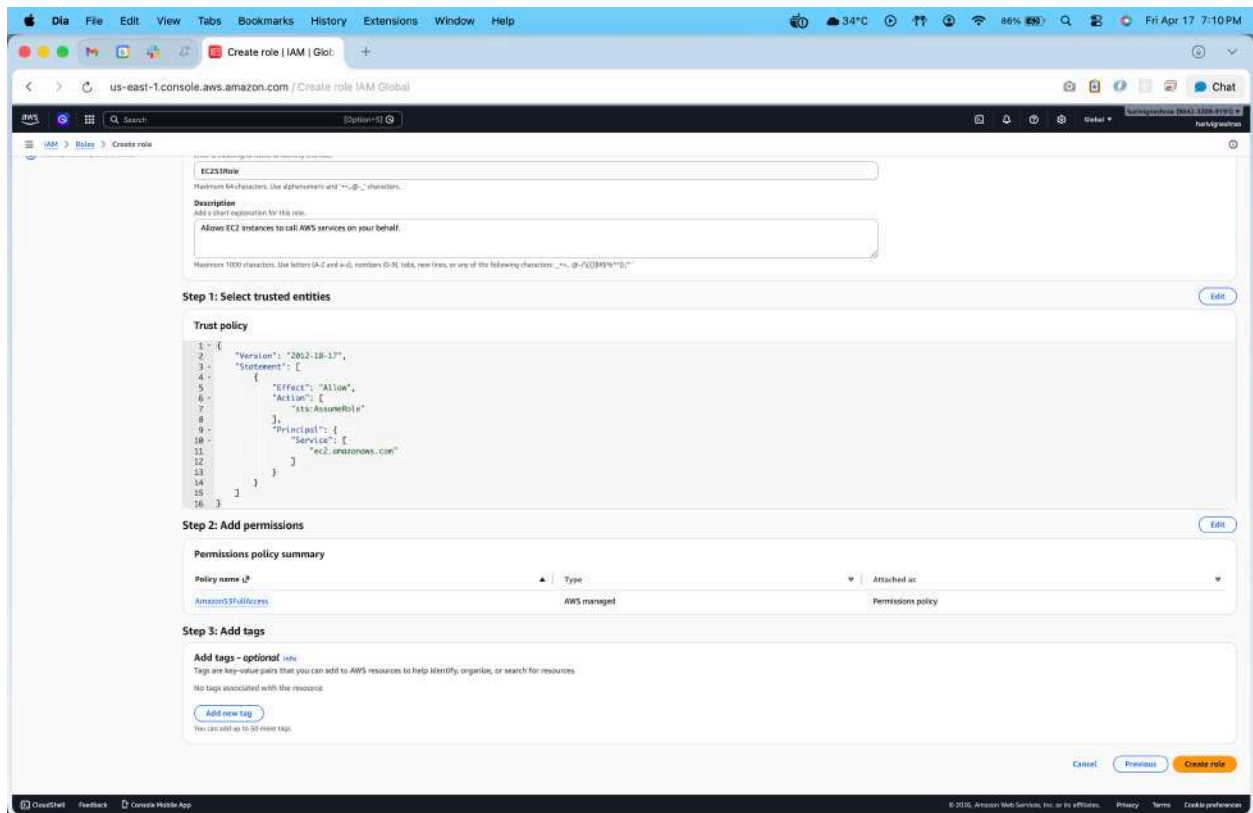
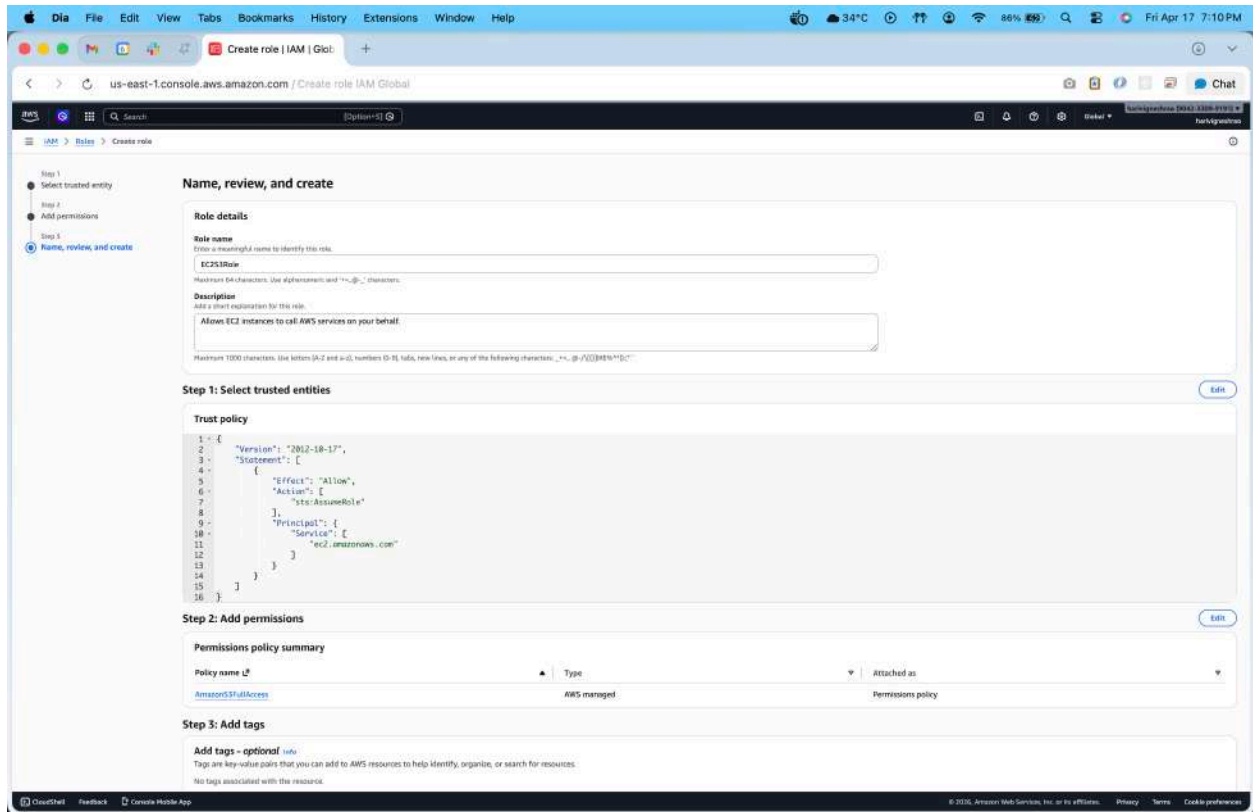
Step 2 — Add Permissions

1. Search policy: AmazonS3FullAccess
2. Select it → Next



Step 3 — Name the Role

- Name: EC2-S3-Role
- Click Create role



Step 4 — Attach Role to EC2 Instance

1. Go to EC2 Dashboard
2. Select running instance
3. Actions → Security → Modify IAM role
4. Select EC2-S3-Role → Update

The screenshot shows the AWS Management Console for the 'ap-south-1' region. The left sidebar contains navigation links for EC2, including Dashboard, AWS Global View, Events, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Capacity Manager, Images, AMIs, AMI Catalog, Elastic Block Store, Volumes, Snapshots, Lifecycle Manager, Network & Security, Security Groups, and Elastic IPs. The main content area displays a list of instances. One instance, 'web' (ID: i-0d294b9e154a4cf15), is selected. The 'Actions' menu is open, showing options like Instance diagnostics, Instance settings, Networking, Security (highlighted), Image and templates, Storage, and Monitor and troubleshoot. The 'Security' option is further expanded, showing 'Change security groups', 'Modify IAM role' (highlighted), 'Get Windows password', 'Image and templates', 'Storage', and 'Monitor and troubleshoot'. Below the instance list, the details for the selected instance are shown, including its state (Running), public and private IP addresses, and DNS information.

Name	Instance ID	Instance state	Instance type	Status check	Alarm
web	i-0d294b9e154a4cf15	Running	t3.micro	OK	

Instance summary

Instance ID	Public IPv4 address	Private IPv4 addresses
i-0d294b9e154a4cf15	13.232.101.112 open address	172.31.40.202

Instance state

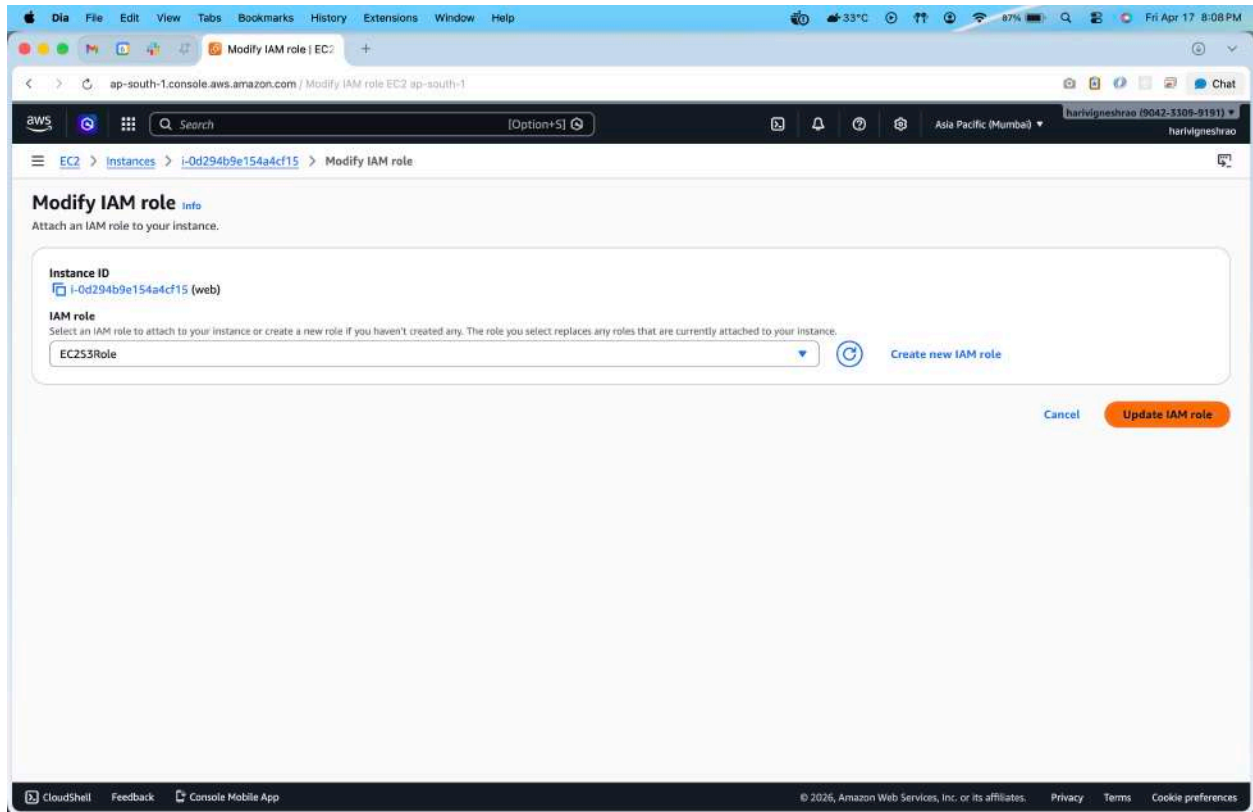
Running

Public DNS

ec2-13-232-101-112.ap-south-1.compute.amazonaws.com | open address

Private IP DNS name (IPv4 only)

ip-172-31-40-202.ap-south-1.compute.internal



Step 5 — Verify from Instance (No Access Keys!)

SSH into instance and run:

```
aws s3 ls
```

```
aws ec2 describe-instances
```

- aws s3 ls Works
- aws ec2 describe-instances **Unauthorized (no EC2 permission)**

Reason: Instance receives **temporary credentials** from IAM Role automatically.

Terminal Shell Edit View Window Help

Connect to instance | +

ap-south-1.console.aws.amazon.com / Connect to instance EC2 ap-south-1

aws [Option+S]

Asia Pacific (Mumbai) harivigneshrao (9042-3309-9191) harivigneshrao

EC2 > Instances > i-0d294b9e154a4cf15 > Connect to instance

Connect Info

Connect to an instance using the browser-based client.

Instance ID
i-0d294b9e154a4cf15 (web)

EC2 Instance Connect SSM Session Manager

Instance ID
i-0d294b9e154a4cf15 (web)

1. Open an SSH client.
2. Locate your private key file. The key used to launch the instance.
3. Run this command, if necessary, to ensure your key file has the correct permissions:
chmod 400 "hero.pem"
4. Connect to your instance using its Public DNS:
ec2-13-232-101-112.ap-south-1.compute.amazonaws.com

✓ Command copied

ssh -i "hero.pem" ubuntu@ec2-13-232-101-112.ap-south-1.compute.amazonaws.com

Note: In most cases, the guessed username is correct.

Cancel

Downloads — ubuntu@ip-172-31-40-202: ~ — ssh -i hero.pem ubuntu@ec2-13-232-101-112.ap-south-1.compute.amazonaws.com

```
ubuntu@ip-172-31-40-202:~$ aws s3 ls
ubuntu@ip-172-31-40-202:~$ aws ec2 describe-instances

aws: [ERROR]: An error occurred (UnauthorizedOperation) when calling the DescribeInstances operation: You are not authorized to perform this operation. User: arn:aws:sts::904233099191:assumed-role/EC2SSMRole/i-0d294b9e154a4cf15 is not authorized to perform: ec2:DescribeInstances because no identity-based policy allows the ec2:DescribeInstances action
ubuntu@ip-172-31-40-202:~$
```

CloudShell Feedback Console Mobile App

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